

Swiss Institute of Banking and Finance



University of St.Gallen

8. Risk Management

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3,130/4,130 Corporate Finance

Course outline

1. General Information
2. Risk and Return
3. Portfolio Theory and CAPM
4. Corporate Valuation
5. Cost of Capital
6. Corporate Financing and the Issuance of Securities
7. Capital Structure
- 8. Risk Management**
9. Conflicts of Interest and Corporate Governance
10. Market Efficiency and Behavioral (Corporate) Finance
11. Empirical Research in Corporate Finance

Agenda

1. Reasons to hedge

- Saving taxes
- Avoiding financial distress
- Capital needs
- Management compensation
- Strategic and investment decisions
- Climate change as new (financial risk)

2. How to measure a firm's risk or risk exposure

3. Channels to reduce risk:

- Natural hedge
- Futures
- Money market hedge

4. Some empirical findings on risk management

Risk management (and failures thereof) are an important topic in the public press...

WSJ: Executives Report Inadequate Risk Management

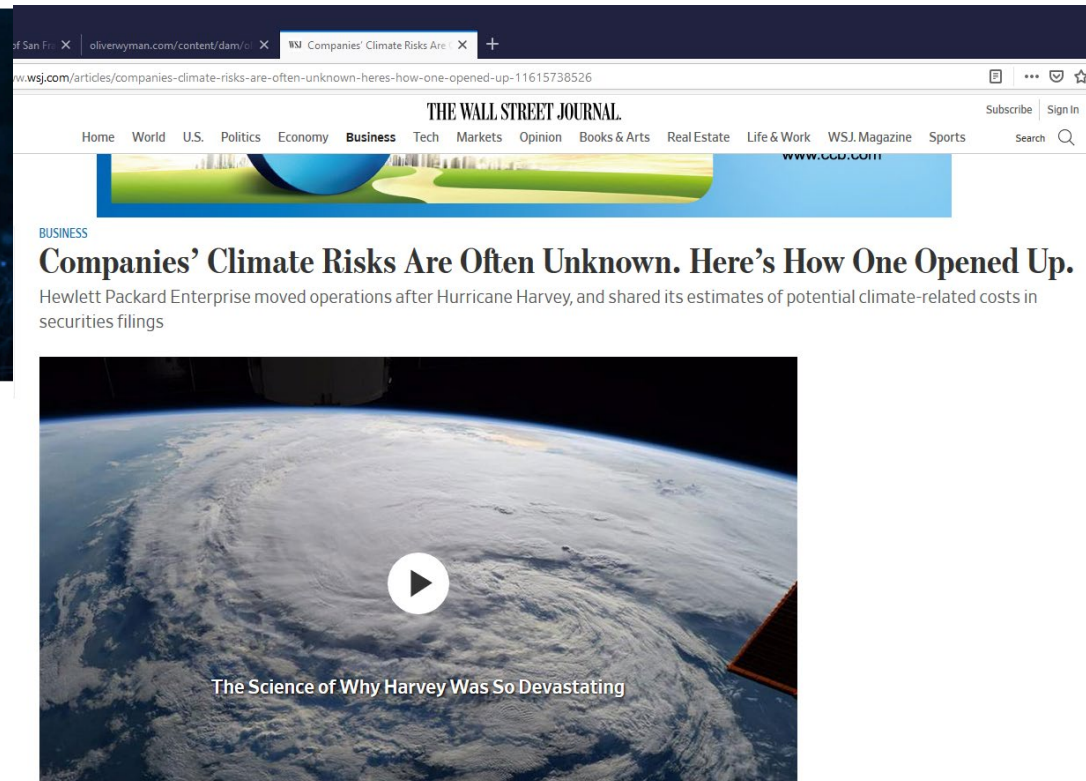
Steven Minsky | Aug. 13, 2014

Despite reports that more than 65% of organizations have adopted Enterprise Risk Management, executives remain unimpressed and skeptical of the value their ERM programs are providing versus what is needed.

A new **report by APQC** finds that fewer than 1 in 5 executives say their companies are effectively managing emerging risk, and the report's authors worry that "Companies may be 'checking the boxes' that say they have processes to monitor strategic risks."**[1]**

Additionally, two in three companies said they did not have a method of ensuring that strategic risks are incorporated into the organization's strategic planning process, and 43% do not have confidence in their method of reporting to the board.

Why the large discrepancy between companies that have adopted ERM and those that are realizing value from ERM implementation? Part of the problem is that **risk managers are spending their time on the wrong things**. The second issue is that board members, unclear about the methods and value of ERM, are wary of empowering their risk managers to take action.



Tropical Storm Harvey's impact on Texas has been catastrophic, but the city of Houston is faring worse than some others. Here's the science of why Harvey has been especially devastating for the area. Photo: NASA

When Hurricane Harvey stalled over Houston and **dumped about 3 feet of rain** in 2017, Hewlett Packard Enterprise Co. 's largest campus was heavily flooded. Its global data center was inundated, along with buildings where the tech company's servers and data-storage gear were assembled.

After the floodwaters receded, Chief Executive Antonio Neri, then the company's newly appointed president, **toured the facility where diesel generators ran machinery drying out rooms to prevent mold**. Two months later, HPE announced it was moving all manufacturing operations out of the Houston area to locations **less exposed to extreme weather**.

Hedging and Modigliani-Miller

MM Proposition I: With perfect capital markets and no taxes the value of the firm is independent of its capital structure.

- It is also possible to apply this theorem to show that, in the absence of market frictions, shareholders are indifferent between hedging on their own accounts and having their firms do the hedging for them.
- If hedging choices do not affect cash flows from real assets, in the absence of taxes and transaction costs, hedging decisions **do not affect firm value.**

Reasons to hedge – Reduction of tax payments

Pretax Income (EBT) for two scenarios			
		p = 50%	p = 50%
		weak Dollar	strong Dollar
Unhedged	EBT	100	-\$20
	Taxes	-40	0
	Income	60	-20
	Expected income = 20		
Hedged	EBT	35	35
	Taxes	-14	-14
	Income	21	21
	Expected income = 21		

Assumption: There is no tax deduction on losses.

Reasons to hedge – Avoiding financial distress

- Financial distress can be costly.
- By hedging risks, a firm can increase its value through a reduction of the probability of facing financial distress in the future.

Example of Microtronics

- Produces scientific equipment (in the U.S.)
- Needs to borrow \$ 100m to refinance an existing loan
- Its operating value depends on the health of the U.S. economy and the dollar/yen exchange rate.
- Strong U.S. economy increases the demand for scientific equipment
- Strong yen increases the dollar cost of Microtronics' competitors

Microtronics' unhedged value in four scenarios

	Strong U.S. economy	Weak U.S. economy
Weak yen	\$ 150m	\$ 95m
Strong yen	\$ 200m	\$ 125m

Reasons to hedge – Avoiding financial distress (II)

Example of Microtronics (cont.)

- Consumers of scientific equipment are reluctant to purchase from a firm with financial difficulties because of potential difficulties in obtaining spare parts.
- The bank is concerned that the recovery of the loan will be significantly lower than \$ 95m in the case of a weak U.S. economy and a weak yen (for calculation of \$ 95m, full consumer confidence is assumed).
- Hedging a part of the currency risk transfers value from the strong to the weak yen scenario (assume that \$ 15m are transferred).

Microtronics' hedged value in four scenarios

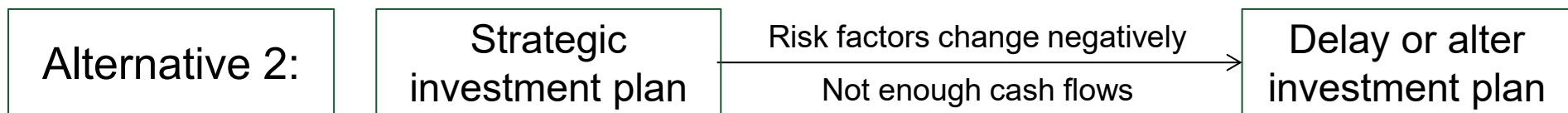
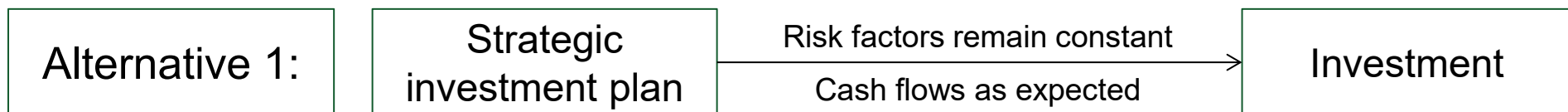
	Strong U.S. economy	Weak U.S. economy
Weak yen	\$ 165m	\$ 110m
Strong yen	\$ 185m	\$ 110m

- Hedging prevents consumers from factoring in a certain probability that Microtronics will be in financial distress.

Reasons to hedge – Capital needs

today future
time →

Decision:



Investors may value the delay or change as a bad signal for firm's future prospects and become overly pessimistic. Therefore, the firm does not get access to external finance (or not at attractive terms).

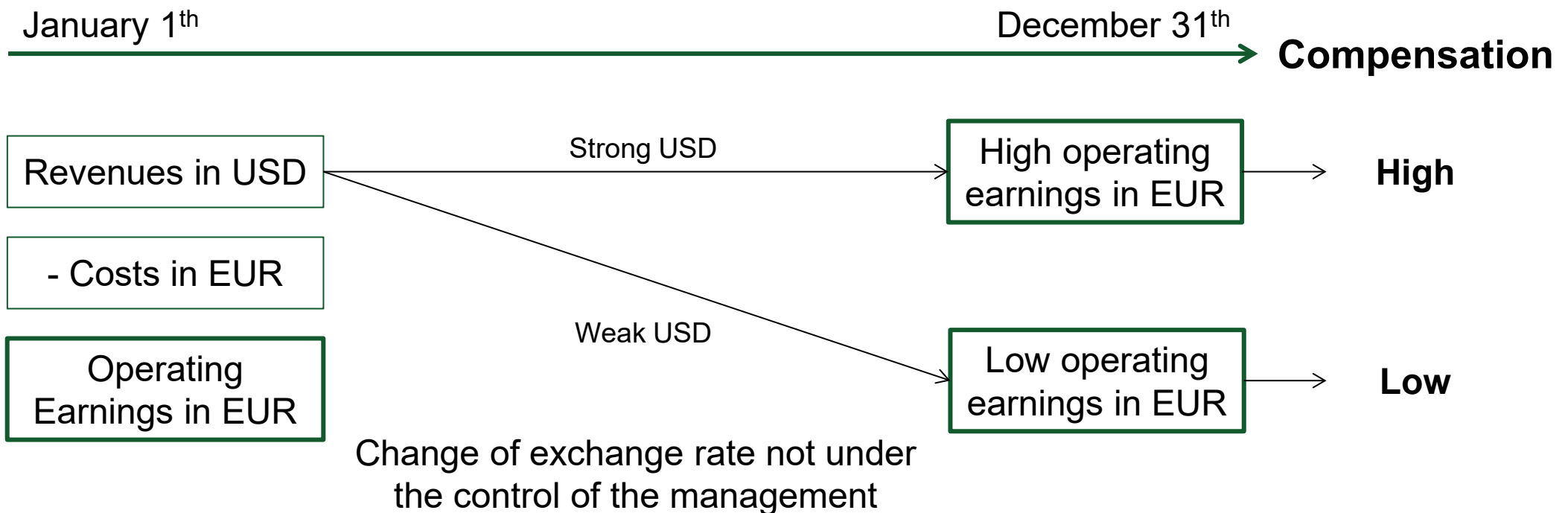
Reasons to hedge – Capital needs (II)

- Internal resources are cheaper than external (e.g., because of information asymmetries, taxes, and transaction costs).
- Therefore, capital expenditures (CAPEX) highly depend on internal cash flows.
- Tendency of overinvestment or underinvestment according to firm's cash flows.
- Problem: Firms have to plan their CAPEX in advance but often face highly volatile cash flows. Hence, they may not have enough cash in the future.
- If a firm cannot undertake or has to alter its strategic investment plan due to changes in the exchange rate (which is just bad luck), investors could be much more pessimistic on the firm's future prospects.
- One possible solution is to hedge cash flows in order to reduce the variability of cash flows, thereby enhancing firm value.

Reasons to hedge – Management compensation

Example:

A firm is based in the Eurozone but sells a significant part of its products in the U.S. The management is compensated based on the operating performance.



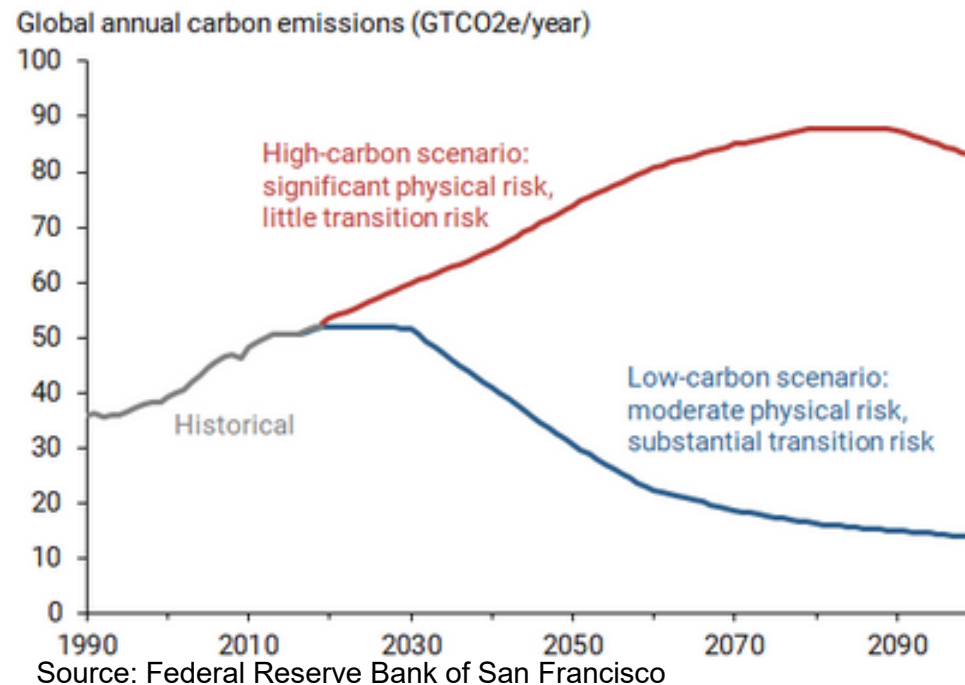
Reasons to hedge – Management compensation (II)

- Managers should only be paid for their «**real**» performance.
- Compensating or punishing the management for performance or risk outside their control is inefficient.
- Hedging uncontrollable risks leaves the managers, especially risk-averse managers, with a larger proportion of manageable risks for performance evaluation.
- This is also desirable from a corporate governance point of view as this reduces the “pay for luck” problem.
- However, it is very difficult to design a compensation package that eliminates a manager’s exposure to all hedgeable risks.
 - Problem of evaluating how such risks affect the firm
 - Problem to find benchmarks that account for multidimensional risks

Reasons to hedge – Strategic & investment decisions

- An active risk management program can improve management's decision-making process by reducing the profit volatility of individual business units.
- Less volatile profits for a company's business units provide management at the firm's central headquarters with better information about where to allocate capital and which managers are the most deserving of promotions.
- Less volatile profits / cash flows also increase debt capacity (and may result in tax shields – see previous lectures).
- A sophisticated risk management will have a better understanding of future market prices so that managers can allocate capital efficiently across the firm's units, because investment decisions rely on future opportunities (e.g., scenarios) and not historical prices.
- Often insurance companies have better abilities to assess risks than industrial companies due to past experiences. Insuring risks of a planned project enables the firm to better price this project.

Reasons to hedge – Climate change as new financial risk



- High-carbon scenario: Rising sea levels, severe storms, floods, and wildfires may disrupt business operations, damage property, and devalue assets.
- Low-carbon scenario: Transformation of the economy to new energy sources and business models; associated with financial risk about the associated climate outcomes and policy responses and the resulting economic and financial fallout.

Measuring a firm's risk exposure: Factor models

$$CF_i = \alpha_i + \beta_{i1}F_1 + \beta_{i2}F_2 + \dots + \beta_{iK}F_K + \varepsilon_i$$

F_K : macroeconomic factors (e.g., interest rate movements, oil price changes)

β_{iK} : cash flows' sensitivity to these factors

$\tilde{\varepsilon}_i$: explanation of the risk not captured in the factors (firm-specific risk)

- A stock's sensitivity to factor risk as well as firm-specific risk is determined by the firm's capital expenditures and operating and financial decisions.
- Factor risk is generally not diversifiable (e.g., exchange rates, oil price), but often it can be hedged by taking off-setting positions in financial derivatives.
- Firm-specific risk is just the opposite (e.g., technical problems of a plant). It is diversifiable but not hedgeable. It is only possible to insure against these risks using insurance contracts.

Measuring a firm's risk exposure: Factor models (II)

- Estimating factor models using regression analysis requires the use of historical data for the risk factors to obtain estimates of the risk factor exposures.
- The dependent variable in these regressions are the unhedged cash flows of the firm.
- Such risk factor exposure estimates will only have predictive power, if the same risk factors affect the firm similarly in the future.
- A simple OLS-based regression approach does not capture non-linear effects of the risk factors on the firm (hence, it may be advisable to also test non-linear specifications).

Measuring a firm's risk exposure: Simulation

- Forward-looking method to estimate risk exposure
- Sometimes useful in rapidly changing industries as regressions estimate the risk factors based on historical data (i.e., are backward-looking)
- But based on strong (distributional) assumptions regarding the relevant risk factors
- Simulations are often used in combination with scenario analysis:
 - Several factors of the model will be held constant (e.g., industry demand)
 - Implementing different scenarios for important factors (e.g., exchange rates) under the condition that all else is constant

Measuring a firm's risk exposure: Factor-based volatility of a firm's cash flows

- Often, it is useful to express the risk exposure of a firm in a single number
- One possibility is to model the risk exposure of the cash flows by using different risk factors (e.g., interest rate, exchange rates, GDP, etc.) and aggregating the risk exposures into a single risk metric.
- In a first step, we regress the cash flows on the identified firm-specific risk factors:

$$CF_i = \alpha_i + \beta_{i1}F_1 + \beta_{i2}F_2 + \dots + \beta_{iK}F_K + \varepsilon_i$$

- In a second step, we can estimate the covariances between each pair of risk factors and calculate the “overall variance of the cash flows”:

$$\sigma^2 = \sum_{m=1}^K \sum_{n=1}^K \beta_m \beta_n \text{cov}(F_m, F_n)$$

- The factor-based volatility of the cash flows is the square root of the variance:

$$\text{Volatility} = \sigma = \sqrt{\sigma^2}$$

Measuring a firm's risk exposure: Factor-based volatility of a firm's cash flows (II)

Example: Computing **factor-based volatility** for a firm's cash flows

Assume that a firm expects a cash flow one year from now (in USDm) that follows the factor model

$$CF = 30 + 2F_{curr} - 4F_{int} + \varepsilon$$

F_{curr} : percentage change in the ¥/USD exchange rate over the next year

F_{int} : percentage change in three-month LIBOR from now until one year from now

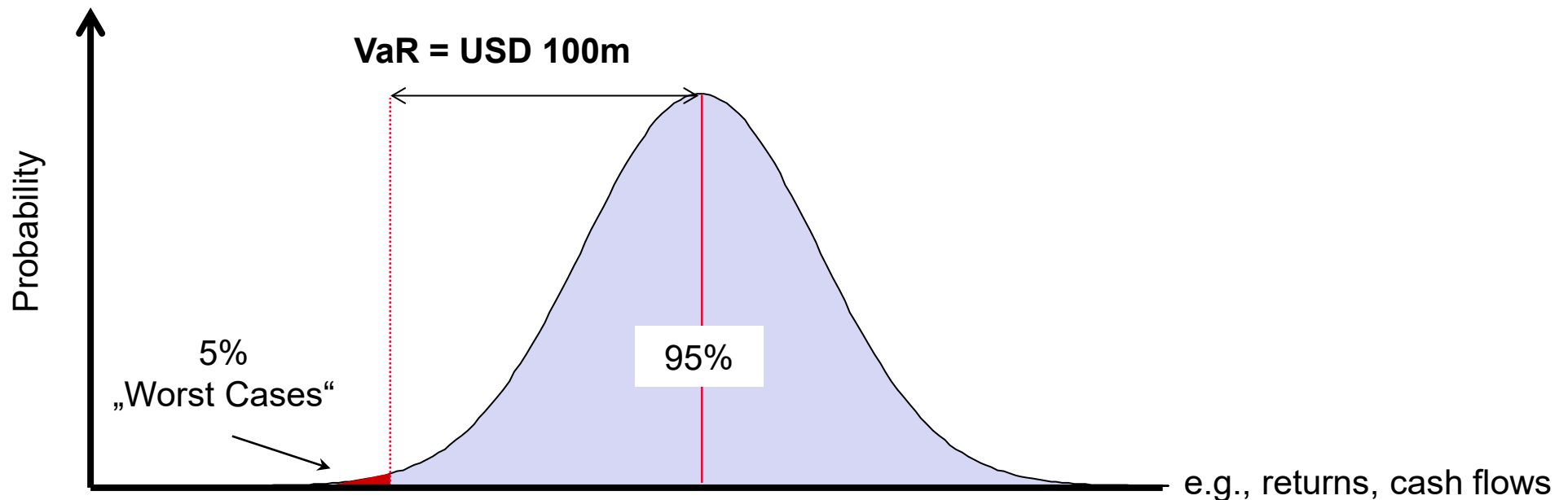
Assume that the variances of the currency and interest factors are estimated to be $\sigma^2_{curr} = 0.011$ and $\sigma^2_{int} = 0.022$, respectively. The covariance between the two factors is $cov_{curr/int} = 0.004$.

$$\sigma_C^2 = 2^2 \cdot 0.011 + 4^2 \cdot 0.022 + 2 \cdot 2 \cdot (-4) \cdot 0.004 = 0.332$$

$$\sigma_C = \sqrt{0.332} = 0.5762 \text{ (in USDm)}$$

Measuring a firm's risk exposure: Value at Risk

- Value at Risk (VaR) is defined as the worst possible loss “under normal market conditions” for a given time horizon.



- For example, a value at risk of USD 100m over the next day at a critical value of 95% means that a potential loss of more than USD 100m within one day has a probability of 5%.

Measuring a firm's risk exposure: Value at Risk (II)

- In general, there are three alternative ways to calculate the VaR:
 - 1) **Parametric VaR:** Based on the assumption of normally distributed returns of all assets (or cash flows of a firm) and therefore also the portfolio consisting of these assets (or divisions). The VaR is exclusively determined by the expected returns or cash flows and the variance-covariance matrix of the individual assets or divisions.
 - 2) **Historical (non-parametric) VaR:** Based exclusively on the historical data (i.e., return / cash flow observations) of the assets or divisions. Advantage: No distributional assumption! Disadvantage: The results strongly depend on the time period used and the approach implies that the past returns or cash flows adequately describe the future returns; in addition, long historical time series are required to obtain meaningful results.
 - 3) **Monte Carlo VaR:** The Monte Carlo approach relies on simulations of the stochastic behavior of returns, cash flows (or risk factors); it requires assumptions regarding the return generating stochastic process and well-specified (possibly non-linear) factor models for the returns.

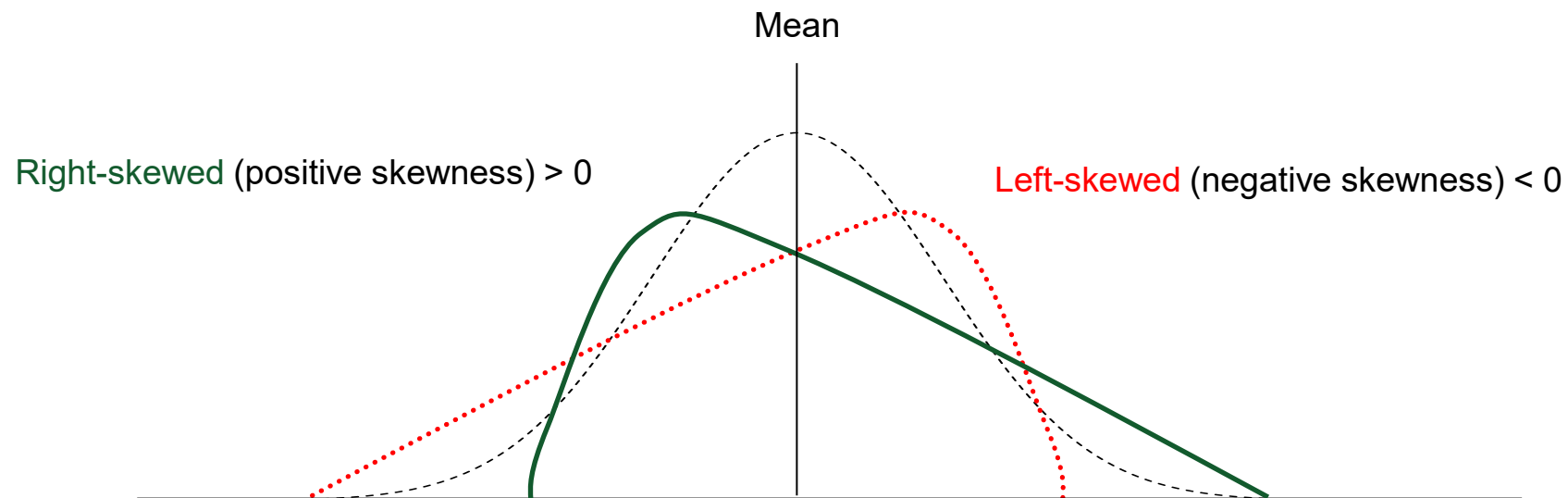
Value at Risk (VaR) – Parametric Value at Risk

- Example: Assume that a firm holds a stock investment with the following characteristics:
 - Investment volume = 10'000'000 USD
 - Daily volatility: $\sigma_{Daily} = 3.5\%$
 - Daily returns are normally distributed
- What is the (parametric) Value at Risk (VaR) of the investment at a 2.5%-level?
$$N(z_c) = 2.5\% \rightarrow z_c \approx 1.96$$
$$VaR(2.5\%) = 10'000'000 \text{ USD} \cdot 1.96 \cdot 0.035 = 686'000 \text{ USD}$$
- What is the 2.5% Value at Risk (VaR) of the investment over a nine day period?
$$Var(2.5\%)^9 = 10'000'000 \cdot 1.96 \cdot 0.035 \cdot \sqrt{9} = 2'058'000 \text{ USD}$$

Value at Risk (VaR) – Problem of parametric VaR

Problem: The distribution of the returns / cash flows is skewed:

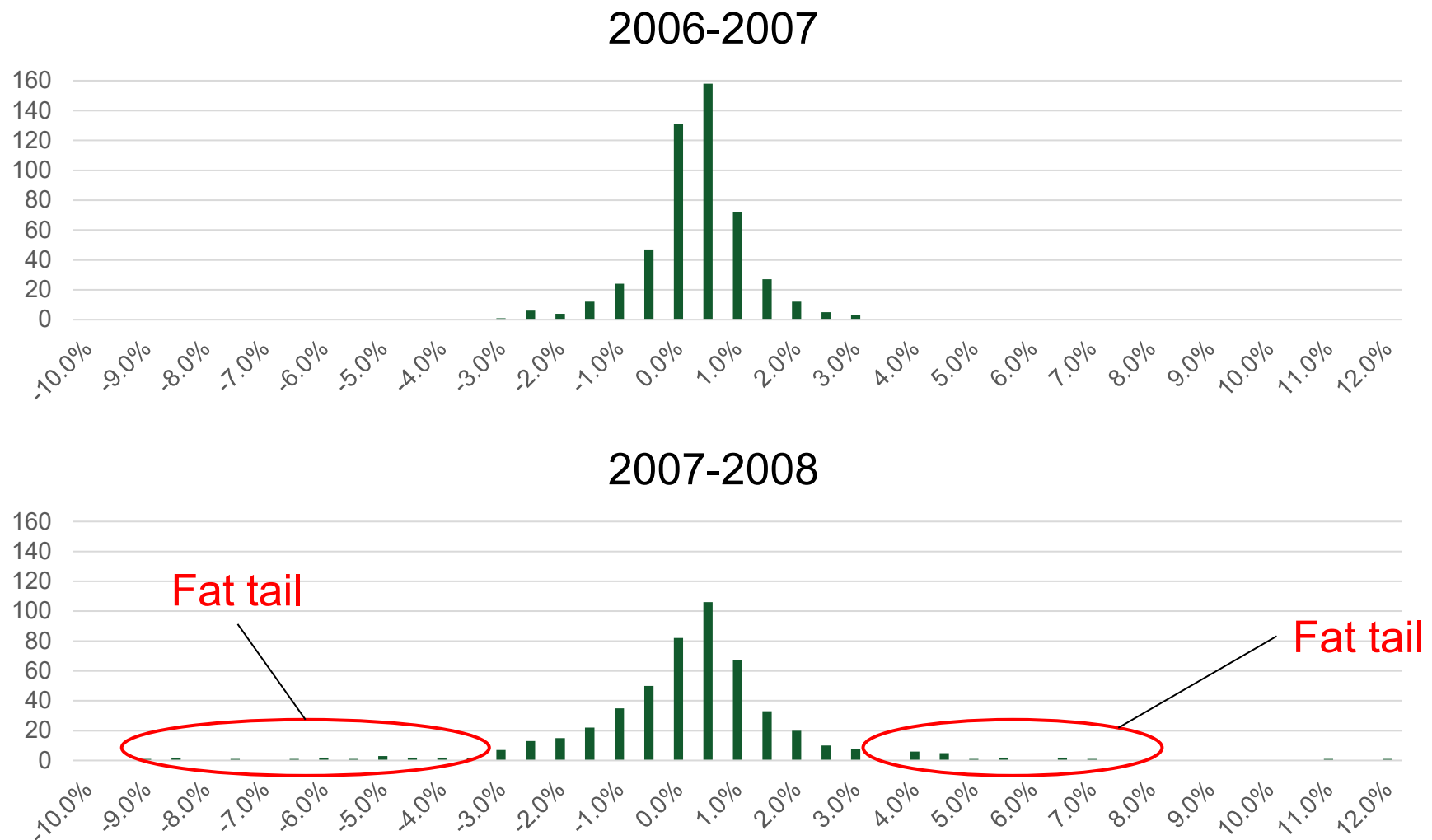
- We generally assume that returns are normally distributed and thereby neglect higher moments (i.e., stocks / assets with the same expected return and volatility have the same VaR).



- Other distributions may better fit a particular underlying: For example, the Weibull distribution is often used to account for fat tails.

Value at Risk (VaR) – Problem of historical VaR

Distribution of daily returns of the S&P 500

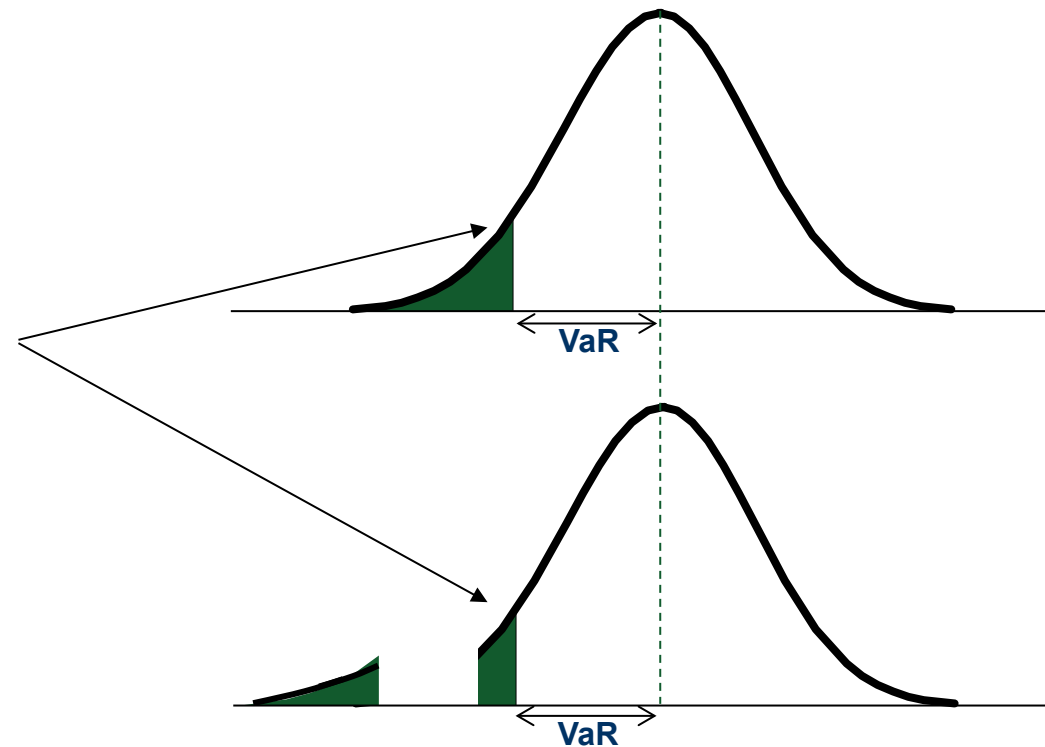


The charts are based on 502 and 504 trading days, respectively.

Value at Risk (VaR) – No information on extreme losses

- Problem: The distribution of the returns / cash flows may comprise extreme values within the tails. The loss within the Value at Risk might be much larger than expected under the assumption of a normal distribution. The expected shortfall can incorporate this problem.

Identical VaR but different expected loss in case the return falls below the VaR value



Measuring a firm's risk exposure: Expected shortfall

- The VaR only considers the frequency of large losses but neglects the magnitude of these losses; this is what the Expected Shortfall does:

$$p\text{-Expected Shortfall} = E(R_i) \text{ for } R_i < (1 - p)\% \text{ percentile}$$

Calculation of a project's Expected Shortfall			
Probability of state	Payoff per state	Investment	Profit
15%	0	-50	-50
10%	10	-50	-40
20%	40	-50	-10
35%	100	-50	50
20%	150	-50	100

Probability	5%	10%	20%	30%	40%
ES _p	-50	-50	-47.5	-40	-32.5

$$ES_{20\%} = \frac{0.15 \cdot (-50) + 0.05 \cdot (-40)}{0.2}$$

$$ES_{20\%} = -47.5$$

Channels to reduce risk – Natural hedge

- Suppose a firm manufactures 2,000 units of a product in Switzerland and sells 1,000 units in Switzerland and 1,000 in the UK. Therefore, the firm is exposed to changes in the exchange rate between GBP and CHF.
- Hence, if the GBP becomes stronger, the firm will earn more money since the production costs in CHF remain stable but the earnings generated in the UK (translated to CHF) increase.

		Strong GBP	Weak GBP
		CHF/GBP 2:1	CHF/GBP 1.5:1
Sales	CH	1000 x 50 CHF = 50'000 CHF	1000 x 50 CHF = 50'000 CHF
	UK	1000 x 40 GBP = 40'000 GBP x 2 CHF / 1 GBP = 80'000 CHF	1000 x 40 GBP = 40'000 GBP x 1.5 CHF / 1 GBP = 60'000 CHF
Production costs		- 90'000 CHF	- 90'000 CHF
EBIT		40'000 CHF	20'000 CHF

Channels to reduce risk – Natural Hedge (II)

- To reduce currency risk the firm can split up the production and therefore the production costs between both currency areas.

		Strong GBP	Weak GBP
		CHF/GBP 2:1	CHF/GBP 1.5:1
CH	Sale	1000 x 50 CHF = 50'000 CHF	1000 x 50 CHF = 50'000 CHF
	Costs	1000 x -45 CHF = - 45'000 CHF	1000 x -45 CHF = - 45'000 CHF
UK	Sale	1000 x 40 GBP = 40'000 GBP	1000 x 40 GBP = 40'000 GBP
	Costs	1000 x -25 GBP = - 25'000 GBP	1000 x -25 GBP = - 25'000 GBP
EBIT CH		5000 CHF	5000 CHF
EBIT UK		15'000 GBP → 30'000 CHF	15'000 GBP → 22'500CHF
EBIT		35'000 CHF	27'500 CHF

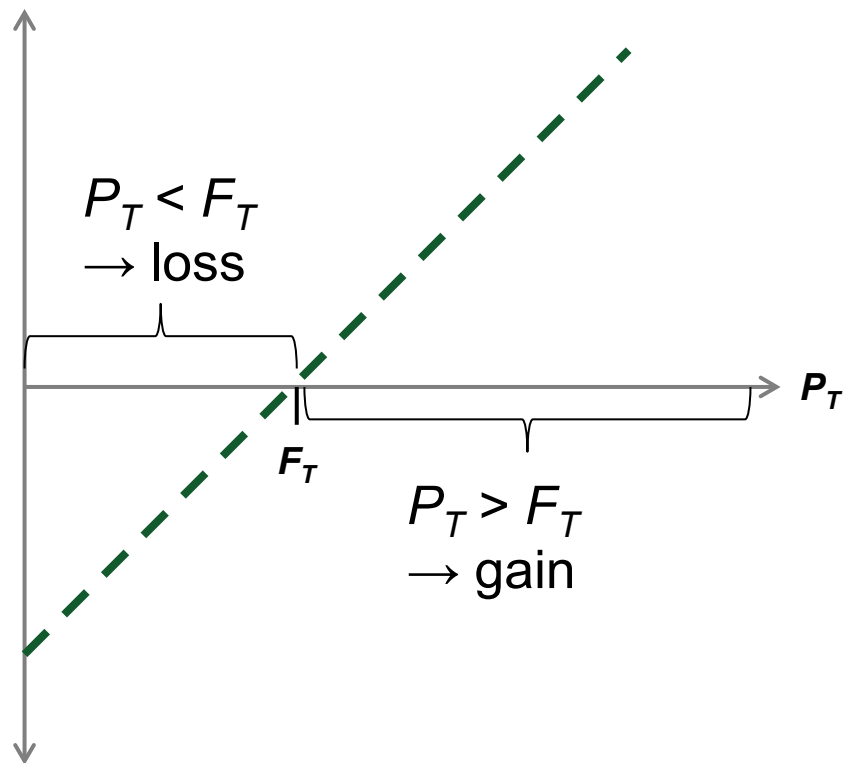
*The costs per unit in the UK are based on an (exogenous) assumption to illustrate the functioning of a natural hedge.

Channels to reduce risk – Futures

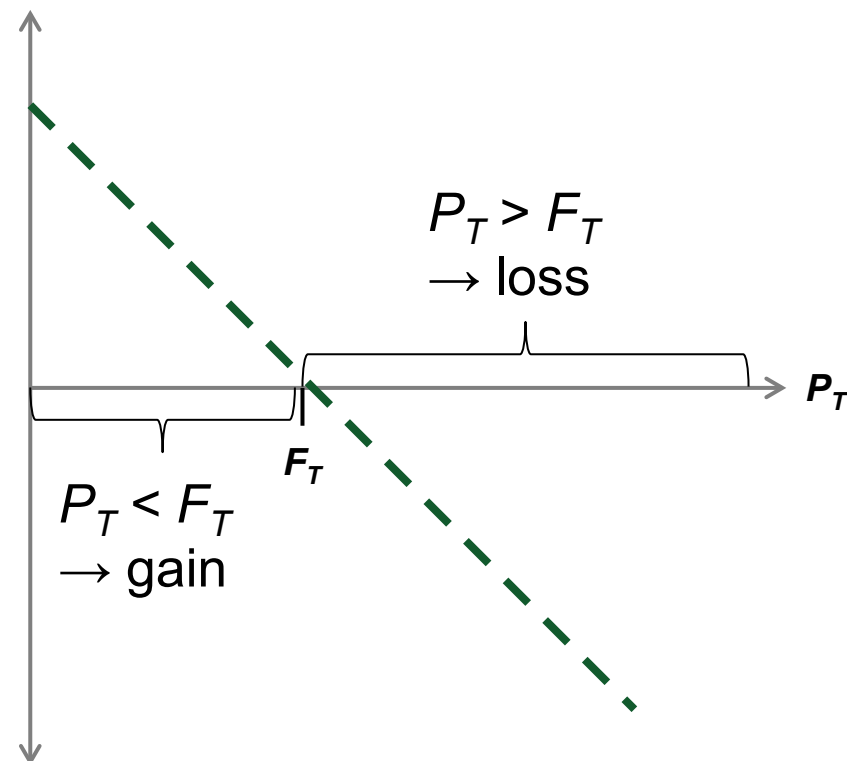
- A forward / future contract is an agreement to buy or sell a security, currency, or commodity at a prespecified price (forward / future price) at some future date.
- A future / forward is a mandatory contract so that both parties have to fulfill their positions. The long position requires the holder to buy the underlying for the future price. Holding the short position obliges to deliver the underlying at the future price.
- In contrast, the spot market is the market for immediate delivery and payment.
- Generally, the forward / future price is set so that the contract is a zero present value investment for both parties ($PV = 0$).
- The payoff structure of a future is linear and therefore symmetric (in contrast to other derivative instruments, such as options).

Channels to reduce risk – Futures (II)

Profit profile of long future



Profit profile of short future

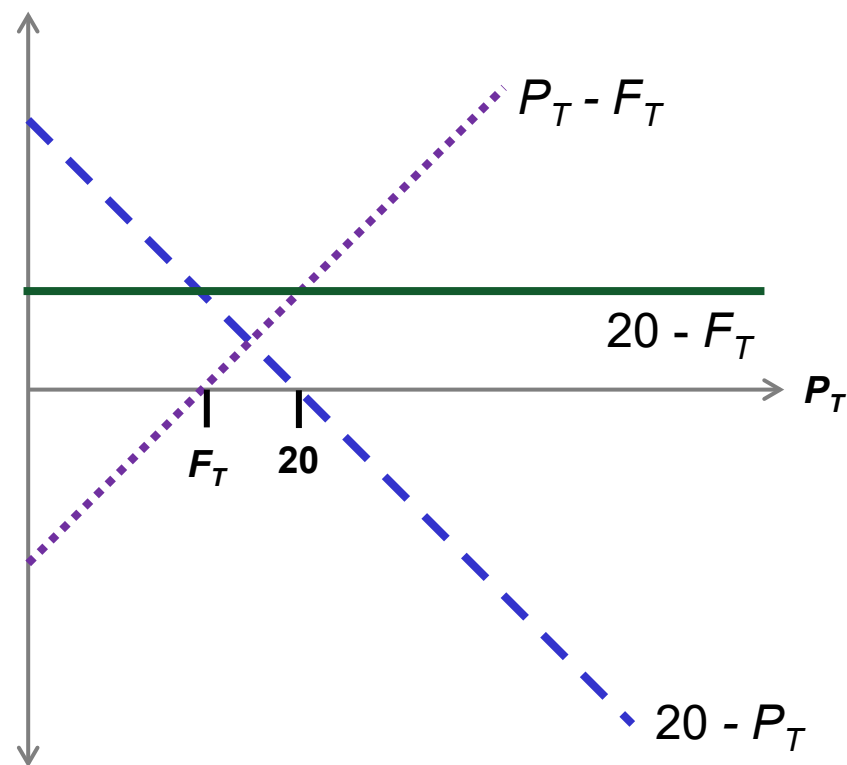


P_T – price at time T

F_T – future price determined today

Channels to reduce risk – Futures (III)

Assume that a refinery has to deliver 1 million barrels of fuel in one year for 20 USD per barrel. At the moment, the single major risk factor is the uncertain purchase price of crude oil. How does hedging with future contracts affect the refinery?



P_T – average purchase price of crude oil

F_T – future purchase price of crude oil

— — — unhedged cash flow from operations

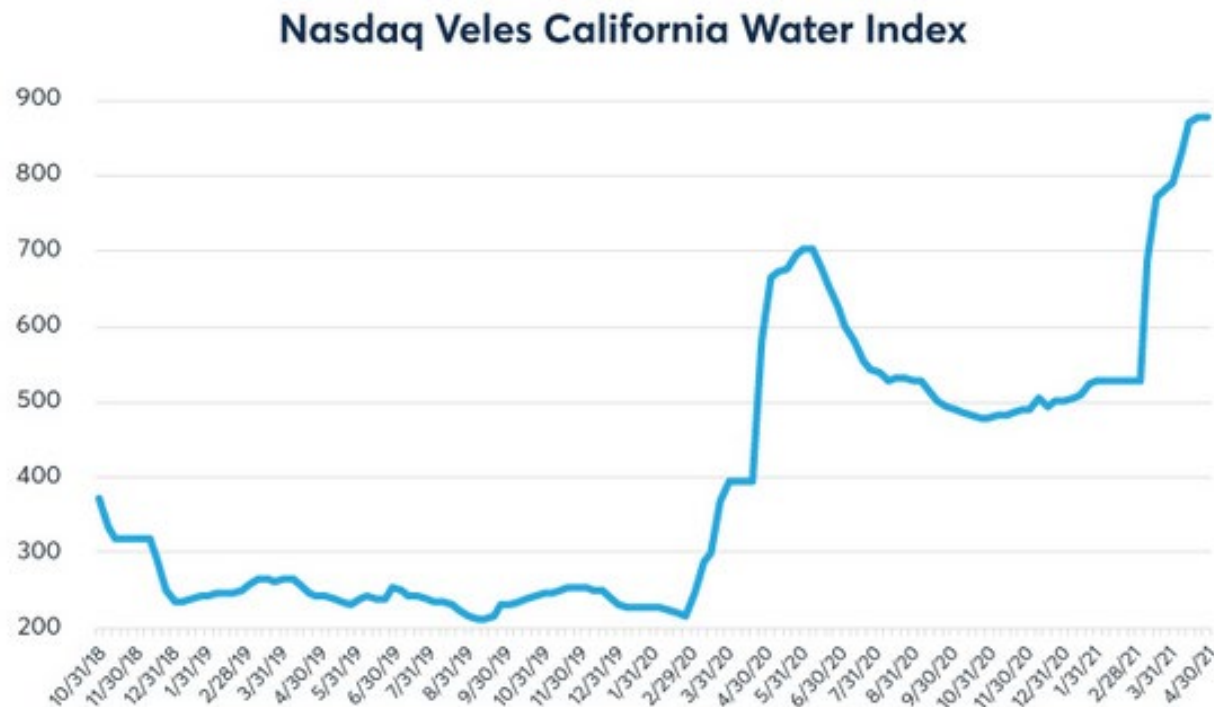
... .. payoff from long future

— — — hedged cash flow from operations

Channels to reduce risk – Futures (IV)

One example of a new instrument to hedge “E”-risk is the Nasdaq Veles California Water Index future, launched in December 2020 by CME Group.

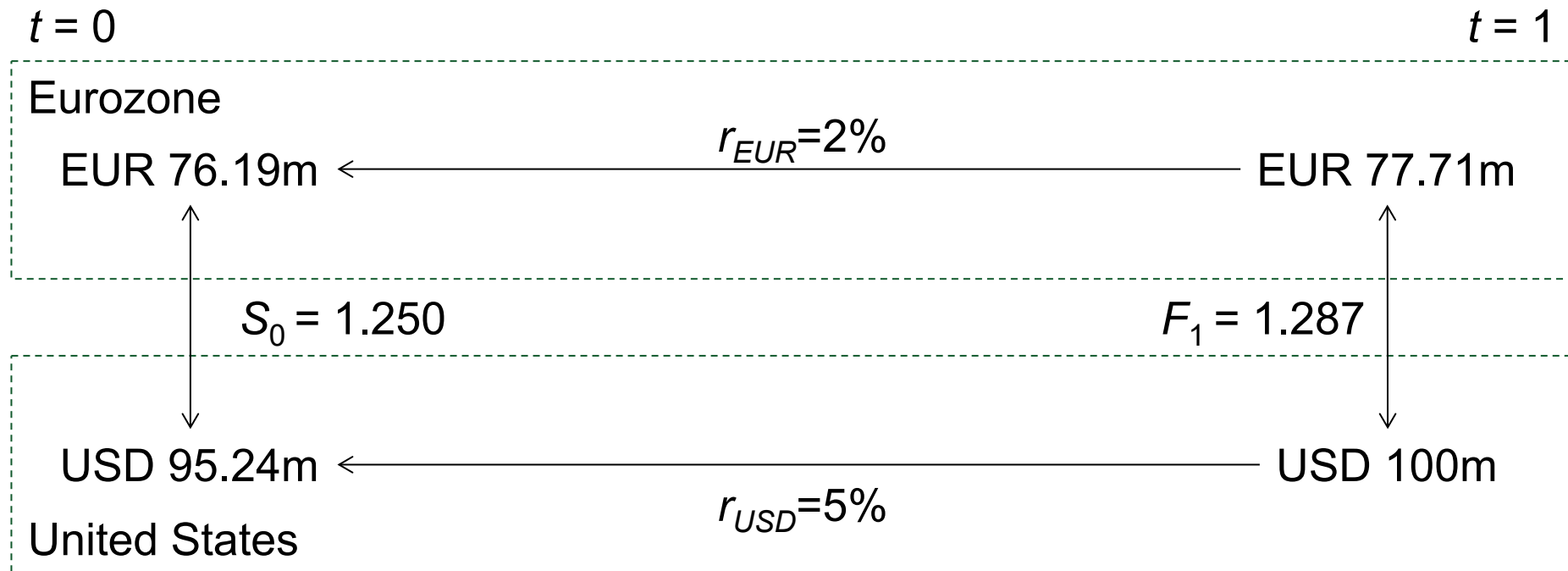
- This futures contract is based on the Nasdaq Veles California Water Index (NQH2O) and allows traders to manage water price risk.



Channels to reduce risk – Money market hedge

- Assume that a company has to pay USD 100m for a production machine in exactly one year. However, the company is based in Europe and all costs and revenues are denominated in Euro. The company would like to hedge this position in the money market.
- The company may exchange EUR into USD today and invest these USD 100m into the machinery in one year.
- A money market hedge involves borrowing one currency on a short-term basis (EUR in the example) and converting it to another currency immediately (USD in the example). In the absence of transaction costs and arbitrage, a money market hedge is exactly like a futures contract.

Channels to reduce risk – Money market hedge (II)



The relation between the spot and future exchange rate can be written as:

$$\frac{F_T}{S_0} = \left(\frac{1 + r_{foreign}}{1 + r_{domestic}} \right)^T \rightarrow \frac{1.287}{1.250} = \left(\frac{1 + 0.05}{1 + 0.02} \right)^1 = 1.029$$

Characteristics of firms which hedge – Examples of empirical findings

Size	Growth opportunities	Leverage
• Larger firms use more derivatives to hedge (Nance, Smith, and Smithson (NSS), 1993) • Inconsistent with lower risk of going bankrupt story. • Smaller firms tend to have higher fix costs to establish a sophisticated risk management. • Larger firms are often only partially hedged and are more exposed to risk factors.	• Firms with more growth opportunities (e.g., high R&D and high M/B) use more derivatives (NSS, 1993). • This ensures to have enough cash flows for investment. • Reduction of the higher financial costs of distress of these firms. • Better allocation of costs on lower taxable earnings (addressing asymmetric tax treatment).	• Graham and Rogers (2002) find “that hedging increases the debt ratio and consequently increases value through the tax benefit of interest deductions.” • If firms with high R&D expenditures and high M/B also have high leverage, they are more likely to hedge.

Empirical findings on risk management

Table 4
Foreign currency derivatives use and firm value: cross-section results

Dependent variable: $\ln$ (Tobin's Q)	All firms with foreign sales > 0	
	Pooled regression (1)	Fixed effects (2)
Observations	2069	2069
R^2	0.73	0.22
FCD dummy (% of Q in parenthesis)	0.053 (5.26%) 2.989***	0.045 (4.53%) 2.273**
Foreign sales/total sales	0.163 4.229***	0.573 5.918***
Size (log of total assets)	-0.071 -7.790***	-0.117 -4.833***
ROA	0.030 11.335***	0.015 11.195***
Debt to equity	0.000 5.004***	0.000 0.237
Growth (capital exp/sales)	0.131 1.367	0.024 0.315
Diversification dummy	-0.102 -4.830***	-0.111 -3.426***
Dividend dummy	-0.090 -3.803***	-0.033 -1.704*
Advertising/assets	1.173 3.879***	0.417 1.622*
R&D/assets	-0.840 -2.330**	-0.418 -0.648

Source: Allayannis and Weston (2001)

- FCD dummy equals 1 if a firm uses currency derivatives and 0 otherwise
- Firms which use currency derivatives have a Tobin's Q^* that is on average 5.3% higher than that of nonusers.

* Tobin's Q = Market value of the firm / replacement cost of assets

Does ESG engagement affect firms' financial risk?

- Hoepner et al. (2024) show that engagement on environmental, social, and governance (ESG) issues can benefit shareholders by reducing firms' downside risk.
- The dataset includes proprietary information on engagement activities by a large UK-based institutional investor.
 - This investor is considered to be one of the most influential activists when it comes to promoting the development of higher ESG standards at portfolio firms.
 - The data include 1,443 engagements across 485 targeted firms worldwide in the period 2005-2018.
- They show that ESG engagement reduces value at risk and lower partial moments, when addressing environmental topics (primarily climate change).
- Target firms with large downside risk reductions exhibit a decrease in environmental incidents after engagement activities.

Key takeaways

1. In a Modigliani-Miller world, risk management does not add any value to a firm.
2. In reality, a sound risk management may increase firm value through several channels (e.g., tax reductions, avoiding costs of distress, etc.).
3. There are different approaches / metrics (e.g., factor model regressions, simulations, VaR) to measure a firm's risk.
4. There is also a menu of tools to hedge a companies main risk exposures, such as natural hedges, derivative contracts, or money market hedges.

Literature

Mandatory readings

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Additional readings

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